Warm up Mini project 2

W1 $P(S) = \frac{6}{20} = 0.10$ $P(H) = \frac{4}{20} = 0.20$ $P(\emptyset) = \frac{5}{20} = 0.25$

W2 $P(S|H) = \frac{3}{4} = 0.75$ $P(S|H^c) = \frac{7}{16} = 0.1875$ 4 times more likely

W3 $P(H|S) = \frac{0.75(0.20)}{0.70} = 0.50$

W4 $P(S) = 0.75(0.20) + 0.1875(0.80)$
$= 0.30$

W5 $P(S \cap H) = \frac{3}{20} = 0.15$ $P(S)P(H) = 0.30(0.20) = 0.06$
Not independent

W6 $P(S|H) = 75\%$ precision $P(H|S) = 50\%$ recall
Misses 50% of all strong cases

W7 $score(S) = 0.30(0.50)\left(\frac{4}{6}\right) = 0.10$ $score(S^c) = 0.70\left(\frac{1}{14}\right)\left(\frac{1}{14}\right) = 0.003571$

$P(S|H,Q) = \frac{0.10}{0.10 + 0.003571} = 0.9655$ compared to 1

$score(S) = 0.30\left(\frac{3}{6}\right)\left(\frac{3}{6}\right) = 0.05$ $score(S^c) = 0.70\left(\frac{13}{14}\right)\left(\frac{13}{14}\right) = 0.603571$

$P(S|H^c, Q^c) = \frac{0.05}{0.05 + 0.603571} = 0.0765$ compared to 0.0769

Both predictions are close to the observed proportions